

$W_1, W_2, \dots, W_n$ subspaces of V

$$\sum W_i = W_1 + W_2 + \dots + W_n = \text{span of } \bigcup_i W_i = \{w_1 + w_2 + \dots + w_n \mid w_i \in W_i\}$$

We say V is an internal direct sum of W_i if any of the following eqv. cond. is true.

(I) $W_1, \dots, W_n$ are lin. indep and span V . (recall $\{W_i\}$ indep means $(w_1 + \dots + w_n = 0 \implies w_i \in W_i) \implies \text{each } w_i = 0$)

$\iff$ (II) $\underbrace{W_1 \times W_2 \times \dots \times W_n}_{V\text{-space via slotwise op}^n} \xrightarrow{s} V$ map given by $s(w_1, w_2, \dots, w_n) = w_1 + \dots + w_n$ is an isomorphism
(Rmk. s automatically linear, surj. if $\sum W_i = V$, injective iff $\{W_i\}$ are indep.)

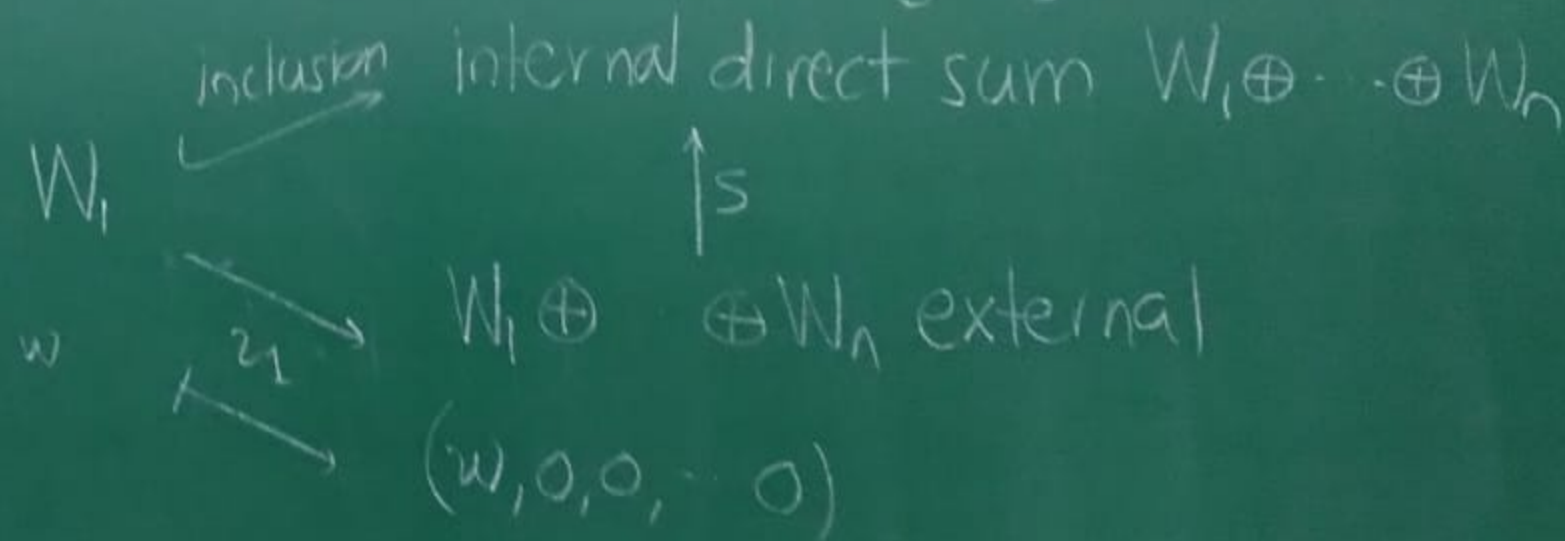
$\iff$ (III) Univ property { Given any vspace U and linear maps $f_i: W_i \rightarrow U$ there exists a
a unique linear $f: W_1 \times \dots \times W_n \rightarrow U$ s.t. $f(0, \dots, 0, w_i, 0, \dots, 0) = f_i(w_i)$

Notation We denote direct sum by $\oplus$ $W_1 \oplus \dots \oplus W_n$ or $\bigoplus_{i=1}^n W_i$.

[Q What happens when we have an infinite family?]

Rmk We have linear maps

Language



Rmk If $\{W_\alpha\}_{\alpha \in I}$ is any family of subspaces of V , then $\bigcap_{\alpha} W_\alpha$ is also a subspace of V .

Spl case $n=2$ When

$V = W_1 \oplus W_2$ we say

" W_2 is a complement of W_1 ."

Thm If W_1, W_2 are subspaces of V then

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$$

Pf 1 Take a basis $\{a_1, \dots, a_k\}$ of $W_1 \cap W_2$. Extend to get a basis of W_1 , also to get one of W_2 etc.

$$W_1 + (v_1, v_2, v_3) = B$$

matrix of f is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Example
looking
ahead

How to find

v_1, v_2, v_3 from A

We will see!

$$\text{Pf 2 } W_1 \oplus W_2 \xrightarrow{s} V$$

external

$$\ker(s) = \{(w, -w) \mid w \in W_1 \cap W_2\} \simeq W_1 \cap W_2$$

$f(v) = Av$ where

$$A = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 2 & 1 \\ -2 & 1 & 2 \end{bmatrix}$$

Calculate $f(v)$ for

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$f(v_1) = v_1 \quad f(v_2) = v_2 \quad f(v_3) = -v_3$$

$$\text{span}(v_1, v_2) = \text{plane } 2x - y - z = 0$$

Linear algebra 0 is valid for vector spaces over any field.
(Informal)

↳ A field is a set with well behaved 4 arithmetic operations

Def: A field F consists of a set F with two binary operations add & mult

s.t. ① $(F, +)$ is an abelian gp with neutral elt 0.

② $(F^{\times}, \cdot)$ (where $F^{\times} = F \setminus \{0\}$) is an abelian gp with neutral elt 1

③ Distributivity holds: $a(b+c) = ab+ac$

Silly stuff

Show $0 \cdot a = 0$ etc

$(w, 0, 0, \dots, 0)$

End of class

$$\mathbb{R}(x) = \left\{ \frac{p(x)}{q(x)} \mid \begin{array}{l} p, q \\ \text{polynomials} \end{array} \right\}$$

Examples of fields: $\mathbb{R}, \mathbb{C}, \mathbb{Q}$

$$\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$$

$$\mathbb{Q}[i] = \{a + bi \mid a, b \in \mathbb{Q}\}$$

$$\mathbb{F}_p = \text{integers mod a prime } p$$